

Carl Qi

Gates Dell Complex
2317 Speedway
Austin, TX, 78712

Website: <https://carl-qi.github.io>
Email: carlq@utexas.edu

Education

University of Texas at Austin <i>Ph.D. in Electrical & Computer Engineering</i>	August 2023 – Present GPA: 3.9
Carnegie Mellon University <i>M.S. in Machine Learning</i>	August 2021 – December 2022 GPA: 3.96
University of California, Berkeley <i>B.A. in Computer Science, B.A. in Applied Mathematics</i>	August 2017 – August 2021 GPA: 3.976 (Summa Cum Laude)

Industry Experience

Amazon Robotics <i>Applied Scientist Intern</i>	Sunnyvale, CA May 2025 – August 2025
• Fine-tuned large vision-language models (VLMs) to enhance robotic failure detection and reasoning • Designed ARMOR, a self-refining algorithm that enables VLMs to iteratively improve their predictions by generating detection outcomes and natural language reasoning conditioned on prior outputs • Achieved state-of-the-art performance with 30% improvement in failure detection rate and up to 100% gain in reasoning accuracy over baseline models; resulted in first-authored paper at ICLR 2026	
UC Berkeley Electrical Engineering and Computer Science (EECS) <i>Instructor CS188 – Intro to AI</i>	Berkeley, CA June 2021 – August 2021
• Gave 25 lectures on fundamental AI techniques such as reinforcement learning to 250+ students • Recruited and led 20 staff members to develop course materials: 2 exams, 5 projects and 10 homework	
Goldman Sachs <i>Quantitative Strategist Intern</i>	New York City, NY July 2020 – August 2020
• Undertook backend development in a digital storefront that delivers cross-asset access to global markets • Developed 2 production-level endpoints that allowed investors to assess risk profile in various scenarios • Integrated the endpoints with an open-source Python library that resulted in 100+ client visits per day	

Research Experience

MIDI Lab, UT Austin <i>Graduate Student Researcher</i>	Austin, Texas August 2023 – Present
• Work with Prof. Amy Zhang on improving generalization of reinforcement learning algorithms	
The Robotics Institute, Carnegie Mellon University <i>Graduate Student Researcher</i>	Pittsburgh, PA August 2021 – June 2023
• Worked with Prof. David Held on computer vision and machine learning for robotic manipulation • Conducted research on policy training and long horizon reasoning for deformable object manipulation • First authored multiple papers in major conferences and got media coverage in the Washington Post	
Berkeley Artificial Intelligence Research (BAIR) <i>Undergraduate Student Researcher</i>	Berkeley, CA April 2020 – April 2021
• Worked with Prof. Pieter Abbeel and Prof. Aditya Grover on robust imitation learning	

Selected Publications

- [1] Carl Qi, Xiaojie Wang, Silong Yong, Stephen Sheng, Huitan Mao, sriram srinivasan, Manikantan Nambi, Amy Zhang, and Yesh Dattatreya. “Self-Refining Vision Language Model for Robotic Failure Detection and Reasoning”. In: *The Fourteenth International Conference on Learning Representations*. 2026. URL: <https://openreview.net/forum?id=jr9hGWQioP>.
- [2] Tal Daniel, Carl Qi, Dan Haramati, Amir Zadeh, Chuan Li, Aviv Tamar, Deepak Pathak, and David Held. “Latent Particle World Models: Self-supervised Object-centric Stochastic Dynamics Modeling”. In: *The Fourteenth International Conference on Learning Representations*. 2026. URL: <https://openreview.net/forum?id=lTaPtGiUUc>.
- [3] Dan Haramati, Carl Qi, Tal Daniel, Amy Zhang, Aviv Tamar, and George Konidaris. “Hierarchical Entity-centric Reinforcement Learning with Factored Subgoal Diffusion”. In: *The Fourteenth International Conference on Learning Representations*. 2026. URL: <https://openreview.net/forum?id=TimC6hxVHj>.
- [4] Carl Qi, Dan Haramati, Tal Daniel, Aviv Tamar, and Amy Zhang. “EC-Diffuser: Multi-Object Manipulation via Entity-Centric Behavior Generation”. In: *The Thirteenth International Conference on Learning Representations*. 2025. URL: <https://openreview.net/forum?id=o3pJU5QCtv>.
- [5] Caleb Chuck, Fan Feng, Carl Qi, Chang Shi, Siddhant Agarwal, Amy Zhang, and Scott Niekum. “Null Counterfactual Factor Interactions for Goal-Conditioned Reinforcement Learning”. In: *The Thirteenth International Conference on Learning Representations*. 2025.
- [6] Caleb Chuck*, Carl Qi*, et al. “Robot Air Hockey: A Manipulation Testbed for Robot Learning with Reinforcement Learning”. In: *CoRR* (2024).
- [7] Carl Qi, Yilin Wu, Lifan Yu, Haoyue Liu, Bowen Jiang, Xingyu Lin, and David Held. “Learning Generalizable Tool-use Skills through Trajectory Generation”. In: *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2024, pages 2847–2854. DOI: 10.1109/IROS58592.2024.10801653.
- [8] Carl Qi, Xingyu Lin, and David Held. “Learning Closed-Loop Dough Manipulation Using a Differentiable Reset Module”. In: *IEEE Robotics and Automation Letters* 7.4 (2022), pages 9857–9864. DOI: 10.1109/LRA.2022.3191239.
- [9] Xingyu Lin*, Carl Qi*, Yunchu Zhang, Yunzhu Li, Zhiao Huang, Katerina Fragkiadaki, Chuang Gan, and David Held. “Planning with Spatial-Temporal Abstraction from Point Clouds for Deformable Object Manipulation”. In: *6th Annual Conference on Robot Learning*. 2022. URL: <https://openreview.net/forum?id=tyxyBj2w4vw>.
- [10] Carl Qi, Pieter Abbeel, and Aditya Grover. “Imitating, fast and slow: Robust learning from demonstrations via decision-time planning”. In: *arXiv preprint arXiv:2204.03597* (2022).

Awards & Honors

Qualcomm Innovation Fellowship Finalist	
Qualcomm	2023
2nd Place Winner of East Coast Regional Datathon	
Citadel Securities	2021
2nd Place Winner in Cisco EN Hackathon	
Cisco	2019

2nd Place Winner in Sodahacks

University of California, Berkeley

2018

1st Prize in Beijing High School Mechanics Contest

Chinese Society of Physics

2015

Teaching

10-418/618: ML for Structured Data

TA

Carnegie Mellon University

Spring 2022

10-725: Convex Optimization

TA

Carnegie Mellon University

Fall 2021

CS188: Artificial Intelligence

Instructor

Univeristy of California, Berkeley

Summer 2021

CS188: Artificial Intelligence

TA

Univeristy of California, Berkeley

Spring 2021

CS188: Artificial Intelligence

TA

Univeristy of California, Berkeley

Fall 2020

CS188: Artificial Intelligence

TA

Univeristy of California, Berkeley

Spring 2020

CS188: Artificial Intelligence

TA

Univeristy of California, Berkeley

Fall 2019

Service

Reviewer

ICLR '25, '26, ICML '26, RSS '25, NeurIPS '25, IROS '24